

46-929, Fall 2019: Homework #4

Due 5:30 PM, Wednesday, November 20, 2019.

Remember that you are to work **individually** on your homework assignments.

1. Do the “R Lab” shown in the attached pages taken from Ruppert, First Edition (Problems 1 thru 4).

Note: In Problem 1, it mentions the use of the “KPSS test.” This is another test for stationarity. You can just use the ADF test solely, or you can read about the KPSS test in the text.

2. Return to the Ford data used on page 38 of Part 6 of the lecture notes. Now consider the daily trading volume.
 - (a) Plot the volume time series, without any transformation. Why would this be a good case to consider using a log transformation of the series?
 - (b) Plot the log Volume series. Do you believe that this series is stationary? Explain. Does taking the first differences help?
 - (c) Fit a time series model to the first-differenced log volume series. In particular, take the following steps:
 - i. Use `auto.arima()` and find the AICC-optimal ARIMA fit to this series. Use `approx=F` and `step=F`. Show the estimated parameters of this model.
 - ii. Inspect the ACF of the residuals of the above fit, and the ACF of the squared residuals from the above fit. What does each tell you?
 - iii. Combine the above ARIMA model with a GARCH(1,1) fit. Show the estimated parameters of this model.
 - iv. Does AIC imply that including the GARCH(1,1) component in the model was useful? Explain.
 - v. Look at the residuals from the ARIMA/GARCH fit above. Is the normal assumption justified?
 - vi. Replace the GARCH(1,1) model with an APARCH(1,1) model. Does AIC justify this additional complexity?

- Hsieh, K. C., and Ritchken, P. (2000) An empirical comparison of GARCH option pricing models. working paper.
- Jarrow, R. (1998) *Volatility: New Estimation Techniques for Pricing Derivatives*, Risk Books, London. (This is a collection of articles, many on GARCH models or on stochastic volatility models, which are related to GARCH models.)
- Pindyck, R. S. and Rubinfeld, D. L. (1998) *Econometric Models and Economic Forecasts*, Irwin/McGraw Hill, Boston.
- Ritchken, P. and Trevor, R. (1999) Pricing options under generalized GARCH and stochastic volatility processes. *Journal of Finance*, **54**, 377–402.
- Rossi, P. E. (1996) *Modelling Stock Market Volatility*, Academic Press, San Diego.
- Tsay, R. S. (2005) *Analysis of Financial Time Series*, 2nd ed., Wiley, New York.

18.16 R Lab

18.16.1 Fitting GARCH Models

Run the following code to load the data set `Tbrate`, which has three variables: the 91-day T-bill rate, the log of real GDP, and the inflation rate. In this lab you will use only the T-bill rate.

```
data(Tbrate, package="Ecdat")
library(tseries)
library(fGarch)
# r = the 91-day treasury bill rate
# y = the log of real GDP
# pi = the inflation rate
Tbill = Tbrate[,1]
Del.Tbill = diff(Tbill)
```

Problem 1 *Plot both `Tbill` and `Del.Tbill`. Use both time series and ACF plots. Also, perform ADF and KPSS tests on both series. Which series do you think are stationary? Why? What types of heteroskedasticity can you see in the `Del.Tbill` series?*

In the following code, the variable `Tbill` can be used if you believe that series is stationary. Otherwise, replace `Tbill` by `Del.Tbill`. This code will fit an ARMA/GARCH model to the series.

```
garch.model.Tbill = garchFit(formula= ~arma(1,0) + garch(1,0), Tbill)
summary(garch.model.Tbill)
garch.model.Tbill@fit$matcoef
```

- Problem 2** (a) Which ARMA/GARCH model is being fit? Write down the model using the same parameter names as in the R output.
 (b) What are the estimates of each of the parameters in the model?

Next, plot the residuals (ordinary or raw) and standardized residuals in various ways using the code below. The standardized residuals are best for checking the model, but the residuals are useful to see if there are GARCH effects in the series.

```
res = residuals(garch.model.Tbill)
res_std = res / garch.model.Tbill@sigma.t
par(mfrow=c(2,3))
plot(res)
acf(res)
acf(res^2)
plot(res_std)
acf(res_std)
acf(res_std^2)
```

- Problem 3** (a) Describe what is plotted by `acf(res)`. What, if anything, does the plot tell you about the fit of the model?
 (b) Describe what is plotted by `acf(res^2)`. What, if anything, does the plot tell you about the fit of the model?
 (c) Describe what is plotted by `acf(res_std^2)`. What, if anything, does the plot tell you about the fit of the model?
 (d) What is contained in the the variable `garch.model.Tbill@sigma.t`?
 (e) Is there anything noteworthy in the plot produced by the code `plot(res_std)`?

Problem 4 Now find an ARMA/GARCH model for the series `del.log.-tbill`, which we will define as `diff(log(Tbill))`. Do you see any advantages of working with the differences of the logarithms of the T-bill rate, rather than with the difference of `Tbill` as was done earlier?

18.17 Exercises

1. Let Z have an $N(0, 1)$ distribution. Show that

$$E(|Z|) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} |z| e^{-z^2/2} dz = 2 \int_0^{\infty} \frac{1}{\sqrt{2\pi}} z e^{-z^2/2} dz = \sqrt{\frac{2}{\pi}}.$$

Hint: $\frac{d}{dz} e^{-z^2/2} = -z e^{-z^2/2}$.